

~~\\i sql/projects_database.sql~~ Installing and Operating the CS345 DevContainer

Contents

- [Installation](#)
- [Install on mac osx](#)
- [Install on Windows](#)
- [The 345 Dev Container](#)
- [Open and build the container](#)
- [To close and exit the dev container](#)
- [To rebuild the dev container](#)
- [Deleting the Docker Containers](#)
- [Working on the Lab using the VSCode interface](#)

Overview

You will be using three software applications in this course:

- PostgreSQL (postgres) a relational database management system and server.
- MongoDB: a non-relational, document-based, database system.
- Python: run apps that connect to relational databases with the psycopg2 library.

In order to simplify the installation and operation of the above, we provide a container that you will install and run on your machine. You will use VSCode (Visual Studio Code) to operate the container, called a “dev container”. The container uses the linux operating system to run the various libraries and resources needed for our course. Containers are commonly used in professional and academic environments to manage software development for data science and data engineering tasks.

Note: You may have python installed on your machine. The container will run its own version of python, independently from the python on your machine. The

same holds true if you have postgres and/or mongodb installed. The container is isolated from your other apps, so you will run all of the 345 software from the dev container using VSCode.

Installation

First, make sure you install any updates to your operating system so you are running the current version. Some software may not run on older versions of some operating systems. It is best practice to keep your system up to date.

1) Install VSCode

If you do not have VSCode installed, do that now.

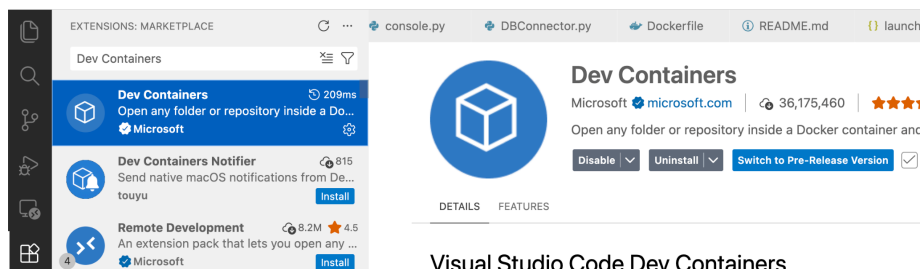
<https://code.visualstudio.com/download>

For Mac use the .dmg file, for Windows use the installer.

There are many good tutorials on installing and running VSCode online. Make sure to include your operating system when searching, e.g. windows 11, osx 26, linux (name of your distro).

Install the dev container extension in VS Code:

- Open the Extensions panel (the “4 boxes” icon).
- Search for Dev Containers
- Install Dev Containers (by Microsoft)



2) Install Docker Desktop

Install on mac osx

1. Download Docker Desktop

- Go to: <https://www.docker.com/products/docker-desktop/>
- Click **Download for Mac**
- Choose the right chip (apple icon upper left/about this mac)
 - **Apple Silicon (M1/M2/M3)** → most newer Macs
 - **Intel** → older Macs

2. Install Docker Desktop

- Open the downloaded .dmg file
- Drag **Docker.app** into the **Applications** folder

3. First Launch & Permissions

- Open **Applications** → **Docker**
- You'll be prompted to:
 - Allow system permissions
 - Enter your Mac password
- Approve anything related to:
 - Networking
 - File access
 - Privileged helper (Docker needs OS access to run on your machine).

4. Confirm It's Running

- Look for the  whale icon in the top menu bar
- Open **Terminal** and run these commands at the prompt:

```
docker --version
```

```
docker compose version
```

You should see versions printed.

Install on Windows

Required for docker desktop:

WSL 2 (Windows Subsystem for Linux 2), allows you to run the linux os used in the container.

Is WSL 2 Required before docker desktop is installed on windows?

If you do not have WSL 2 enabled, the Docker Desktop Installer for Windows will typically attempt to enable and configure the WSL 2 feature for you during the installation process.

While Docker Desktop can automate the setup, you may prefer to manually run `wsl --install` in PowerShell as an administrator prior to installation. This ensures your Windows Subsystem for Linux environment is up-to-date and avoids potential permission or reboot-loop hiccups during the Docker installation phase.

If you are going to use the Docker installer to deal with WSL 2, skip to step 2 below.

Enable WSL (one-time setup)

- Open PowerShell as Administrator
- Run: `wsl --install`

Restart your computer when prompted.

Check in a terminal such as powershell: `wsl --status`

You should see:

Default version: 2

A Linux distro (Ubuntu)

Download and Install Docker Desktop

- Go to: <https://www.docker.com/products/docker-desktop/>
- Click **Download for Windows**

Install:

Run the installer

Important: leave these checked:

- Use WSL 2 instead of Hyper-V
- Add Docker to PATH

Restart when asked.

Start Docker Desktop

- Launch **Docker Desktop**
- It may take a minute or two the first time
- You'll see the  whale icon in the system tray

Confirm the Install

Open: either a Windows Terminal, PowerShell

Run these commands:

```
docker --version
```

```
docker compose version
```

You should see versions printed.

Notes:

- **You do NOT need Docker Hub for 345.**
- You should be able to use power shell terminal windows.

- In case you have issues, an older requirement was to always use **WSL-based terminals** (Ubuntu, VS Code, etc.) Best way to open the WSL terminal: Windows Terminal, select Ubuntu.

3) The 345 Dev Container

Download the file: cs345_dev_container-vXzip (where X is the current version supplied on canvas). We'll use 1.3.1 as the current version example below:

Unzip the file and make sure you have this folder/file configuration (in the file explorer or finder):

```
Downloads/  
    cs345_dev_container-v1.3.1
```

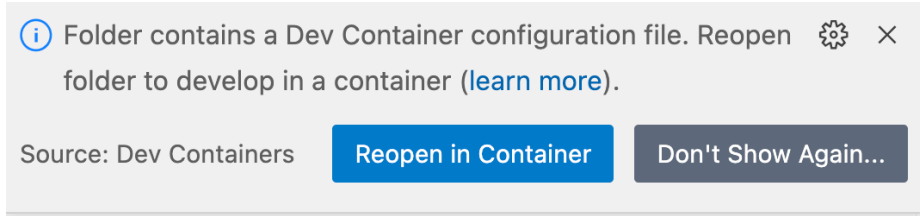
If you have this:

```
Downloads/  
    cs345_dev_container-v1.3.1/  
        cs345_dev_container-v1.3.1
```

Remove the top cs345_dev_container-v1.3.1 so there is only one cs345_dev_container-v1.3.1 folder.

Next, Open the folder with VSCode and build the container :

- Launch VSCode
- Select "Open Folder"
- Navigate to the cs345_dev_container-v1.3.1 folder and select it. VSCode will recognize this is as a dev container. You will see this popup:

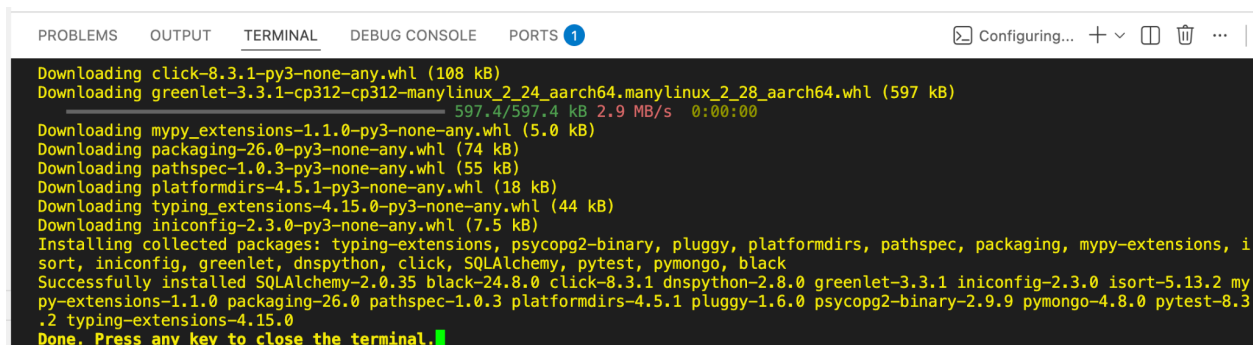


Select “Reopen in Container”.

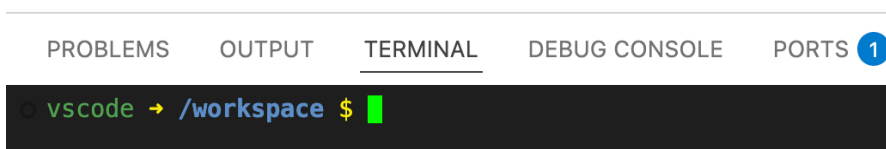
IF you don't get a pop-up to “Reopen in Container”

- Open the VS Code Command Palette.
- macOS: Cmd+Shift+P, Windows: Ctrl+Shift+P
- Search “Dev Containers” and run the command “Dev Containers: Reopen as container”
- This should reload the VS Code window and load in the container

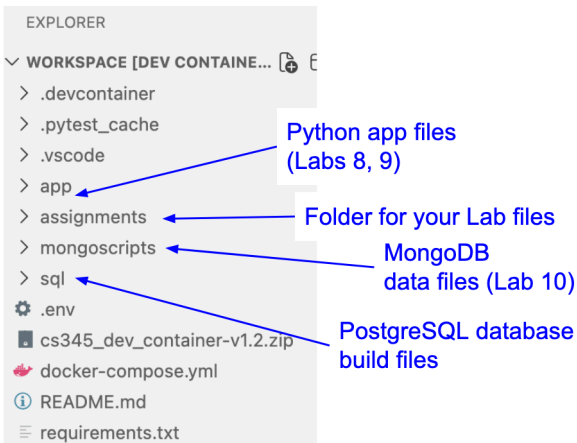
VSCode will build the dev container. This may take a minute. Eventually, you should see this output:

A screenshot of the VS Code terminal window. The terminal shows the process of downloading and installing various Python packages. The output includes: "Downloading click-8.3.1-py3-none-any.whl (108 kB)", "Downloading greenlet-3.3.1-cp312-cp312-manylinux_2_24_aarch64.manylinux_2_28_aarch64.whl (597 kB)", "Downloading mypy_extensions-1.1.0-py3-none-any.whl (5.0 kB)", "Downloading packaging-26.0-py3-none-any.whl (74 kB)", "Downloading pathspec-1.0.3-py3-none-any.whl (55 kB)", "Downloading platformdirs-4.5.1-py3-none-any.whl (18 kB)", "Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)", "Downloading iniconfig-2.3.0-py3-none-any.whl (7.5 kB)", "Installing collected packages: typing-extensions, psycpg2-binary, pluggy, platformdirs, pathspec, packaging, mypy-extensions, i", "sort, iniconfig, greenlet, dnspython, click, SQLAlchemy, pytest, pymongo, black", "Successfully installed SQLAlchemy-2.0.35 black-24.8.0 click-8.3.1 dnspython-2.8.0 greenlet-3.3.1 iniconfig-2.3.0 isort-5.13.2 my", "py-extensions-1.1.0 packaging-26.0 pathspec-1.0.3 platformdirs-4.5.1 pluggy-1.6.0 psycpg2-binary-2.9.9 pymongo-4.8.0 pytest-8.3", ".2 typing-extensions-4.15.0", "Done. Press any key to close the terminal." The terminal has a dark background with yellow and green text.

Click any key in the terminal window to see this prompt:

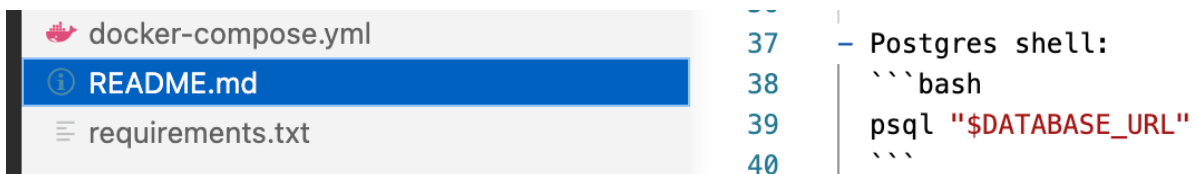
A screenshot of the VS Code terminal window. The terminal shows the prompt "vscode -> /workspace \$" with a green cursor. The terminal has a dark background with light blue text. The tabs at the top are "PROBLEMS", "OUTPUT", "TERMINAL" (selected), "DEBUG CONSOLE", and "PORTS 1".

You see these folders/files in the file explorer in the left-hand pane in VSCode:

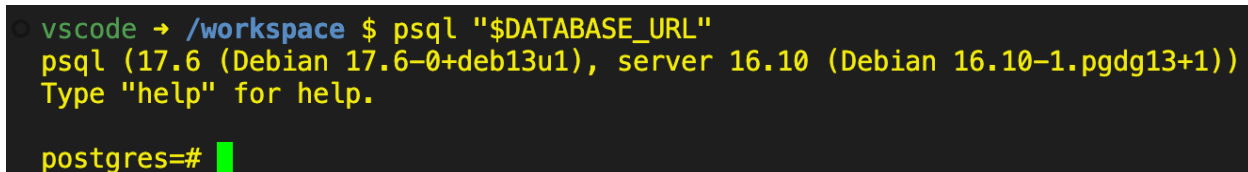


Next, run the psql shell interface that allows access to the PostgreSQL database server:

Open the README file and copy the path variable for postgres:



Paste into the prompt and click Return.



You are now ready to work with postgres. You will use psql for most of the labs in 345.

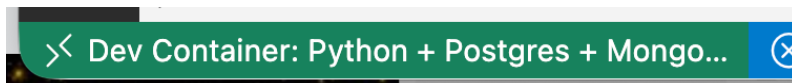
Congratulations! You are now ready to begin Lab 1.

The following notes provide some extra information if needed.

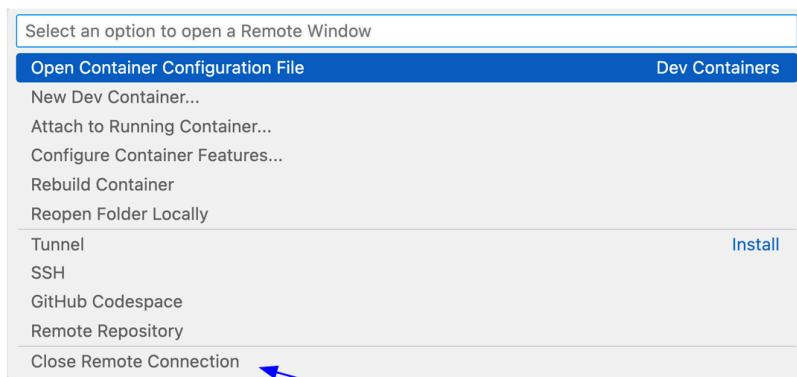
To close and exit the dev container

If you need to use VSCode for other work, you can exit the dev container.

Click on the green area in the lower left corner:



Select **Close Remote Connection** from the command palette drop down that appears:

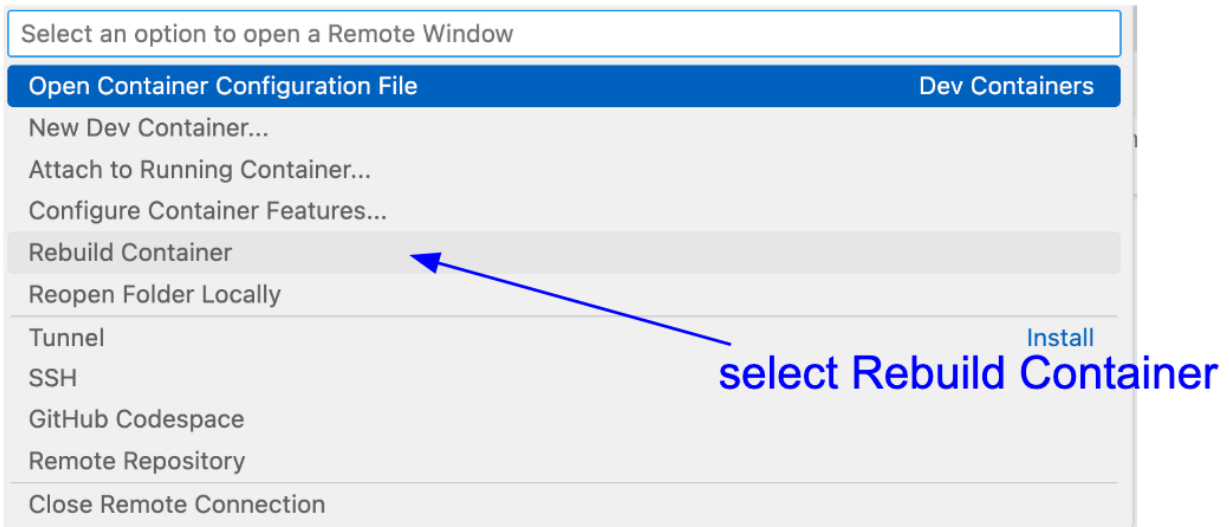


select Close Remote Connection

To return to the 345 dev container, close the current project/folder in VSCode, then open the cs345_dev_container-v1.3.1 folder and click on “Reopen in Container”.

To rebuild the dev container

There won't be any need for this, but if any changes are made to the config files, the dev container will need to be rebuilt. Choose Rebuild Container from the command palette.



Deleting the Docker Containers

The dev container includes three Docker containers (called services). You can see these by opening a terminal window and run `docker ps` to list the running containers:

```
gordon@turnip ~ % docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                    NAMES
cc58e2a29c5d   cs345_dev_container-v12-app         "/bin/sh -c 'echo Co..." 26 minutes ago Up 26 minutes                dev-app
33082b6c480e   postgres:16                         "docker-entrypoint.s..." 26 minutes ago Up 26 minutes (healthy)    5432/tcp          dev-postgres
98460a69fdf6   mongo:7                             "docker-entrypoint.s..." 26 minutes ago Up 26 minutes (healthy)    27017/tcp         dev-mongo
```

In the rare case where you want to rebuild the dev container from the beginning, after closing the remote connection in vscode, you need to: stop all running containers:

To remove all containers:

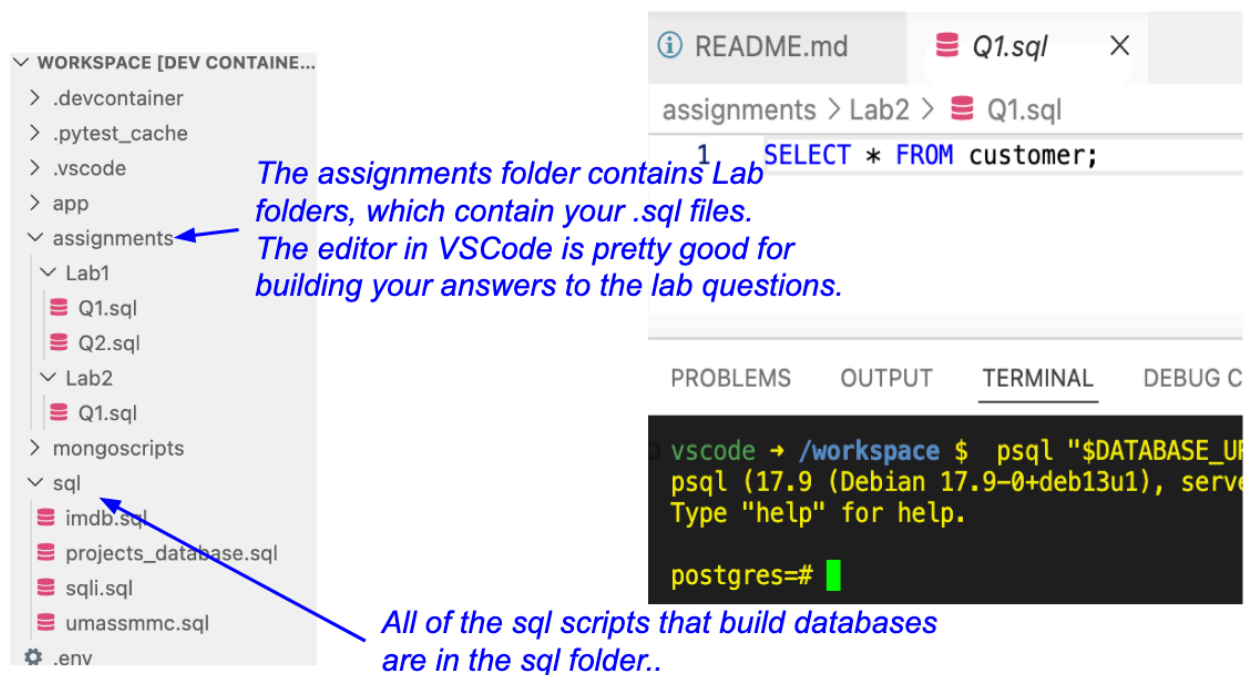
```
docker rm -f dev-mongo dev-postgres dev-app 2>/dev/null || true
```

```
[gordon@cicsmac-us-152 ~ % docker rm -f dev-mongo dev-postgres dev-app 2>/dev/null || true
dev-mongo
dev-postgres
dev-app
[gordon@cicsmac-us-152 ~ % docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                    NAMES
[gordon@cicsmac-us-152 ~ %
```

Now you can rebuild the dev container in VSCode by following the steps in this document.

Working on the Lab using the VSCode interface

We suggest creating folders for each lab and creating your sql files in the corresponding lab folder:



The postgres window is open in the bottom pane (you can also open it as a separate window), and you can use the editor to work on your lab files.